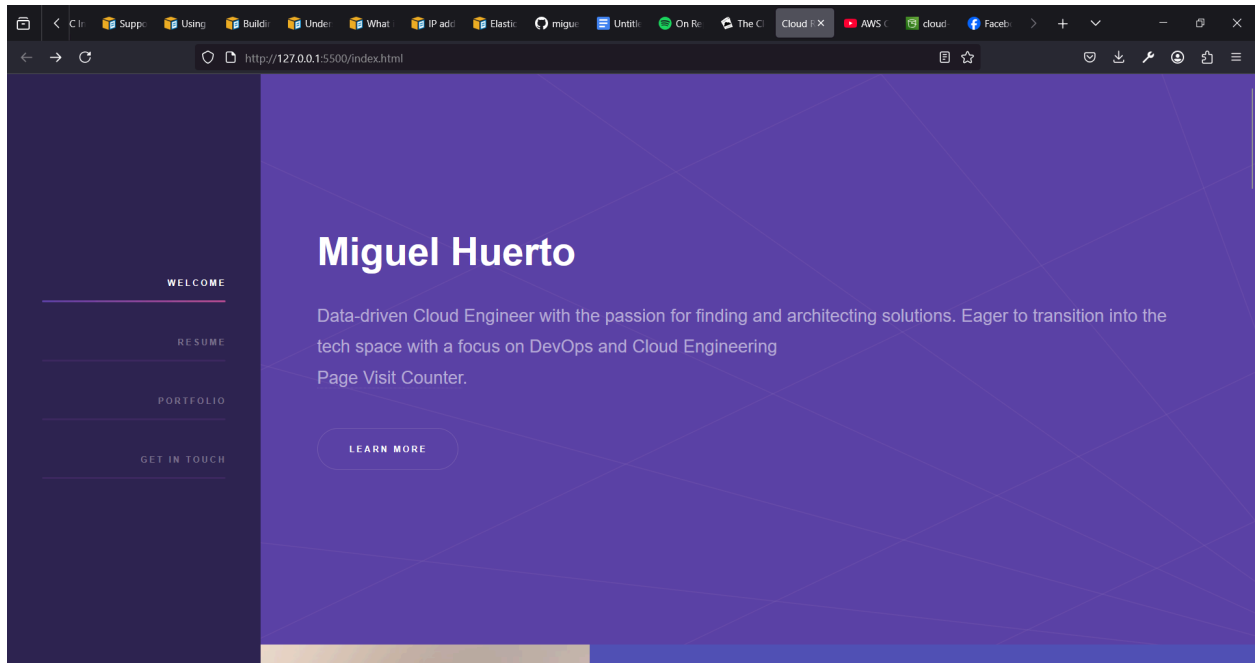


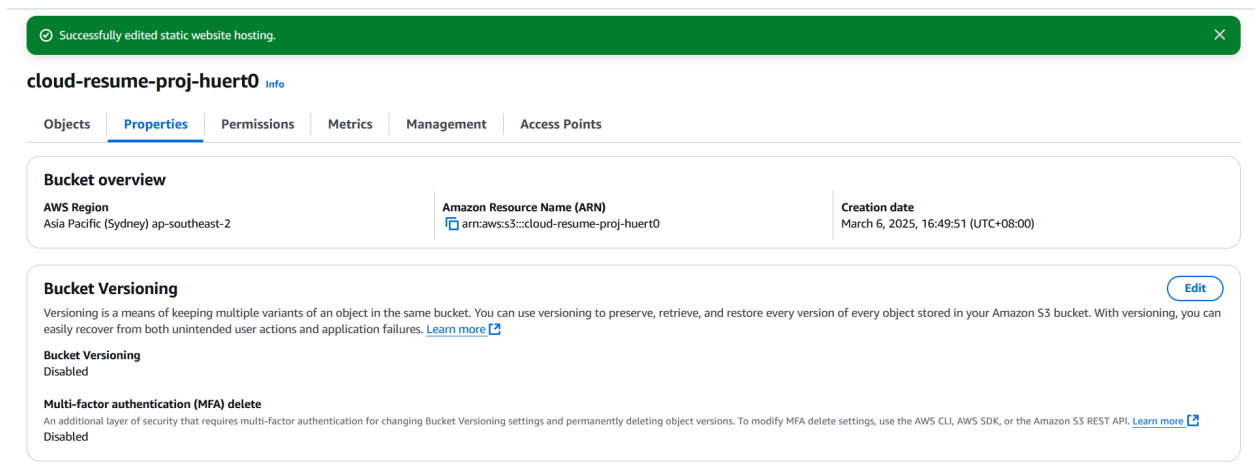
AWS Cloud Resume Challenge

This project will be composed of several parts

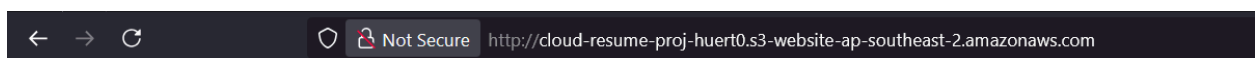
1. HTML&CSS



2. Static Website



Public Access Blocked



403 Forbidden

- Code: AccessDenied
- Message: Access Denied
- RequestId: VMT5QD7DWR5AXDX2
- HostId: Wgvmp5Z3aEDzXLDJjMfZi0ScUrVN3a+aYcuS11dPpLNAeuDQ6RdTztjDYdixEE7vLjv53rG8nV4NWMKbnJcGNsTyER6+it7o

Use CloudFront for low latency high transfer speed delivery of data, CDN.

CloudFront > Distributions > E2YHV001CBLB9

Introducing the CloudFront Security Dashboard
The new security tab is a unified place to configure, manage, and monitor security for your CloudFront distribution. The built-in dashboard gives you visibility into top security trends, allowed and blocked traffic, as well as visibility and controls for bots. CloudFront geographic restrictions are now part of the security dashboard.

Successfully created new distribution.
To get in-depth monitoring information for your distribution's internet traffic, [create an Internet Monitor](#).

The S3 bucket policy needs to be updated
Complete distribution configuration by allowing read access to CloudFront origin access control in your policy statement. [Go to S3 bucket permissions to update policy](#).

Policy statement copied

Copy policy

E2YHV001CBLB9 [View metrics](#)

General

Security

Origins

Behaviors

Error pages

Invalidations

Tags

Logging

Details

Distribution domain name
[d19seotqekwxyr.cloudfront.net](#)

ARN
[arn:aws:cloudfront::084828560751:distribution/E2YHV001CBLB9](#)

Last modified
Deploying

Settings

Description

Alternate domain names

Standard logging

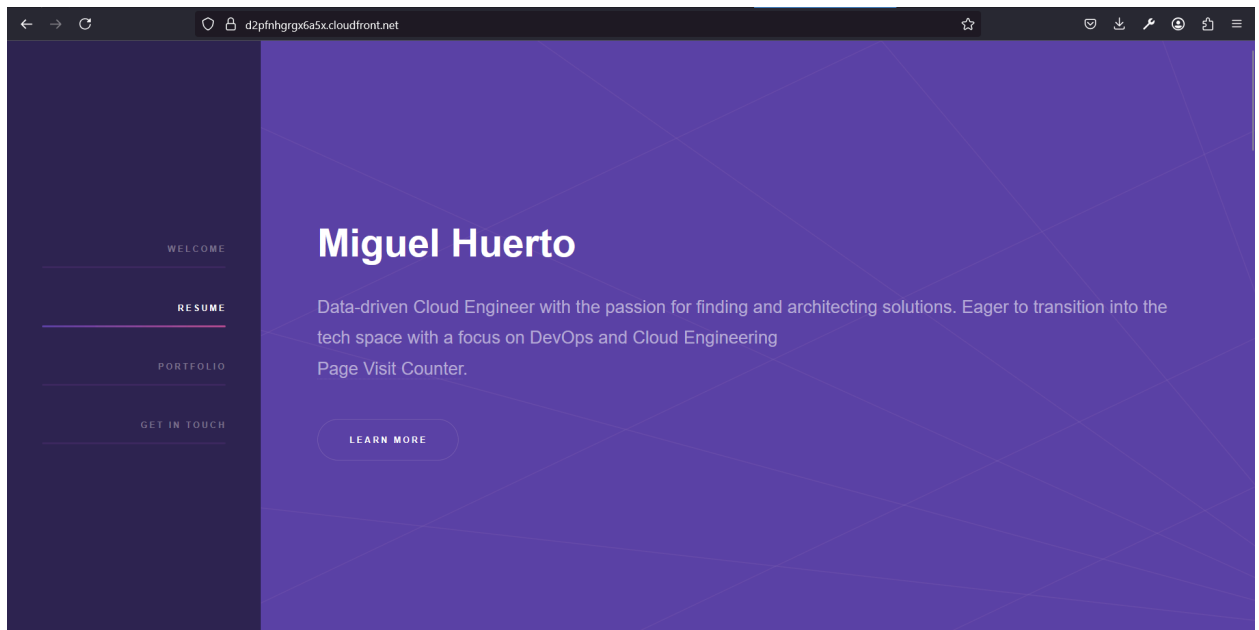
Edit

HTTPS only

Edited Permissions

```
{
  "Version": "2008-10-17",
  "Id": "PolicyForCloudFrontPrivateContent",
  "Statement": [
    {
      "Sid": "AllowCloudFrontServicePrincipal",
      "Effect": "Allow",
      "Principal": {
        "Service": "cloudfront.amazonaws.com"
      },
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::cloud-resume-proj-huert0/*",
      "Condition": {
        "StringEquals": {
          "AWS:SourceArn": "arn:aws:cloudfront::084828560751:distribution/E2YHV001CBLB9"
        }
      }
    }
  ]
}
```

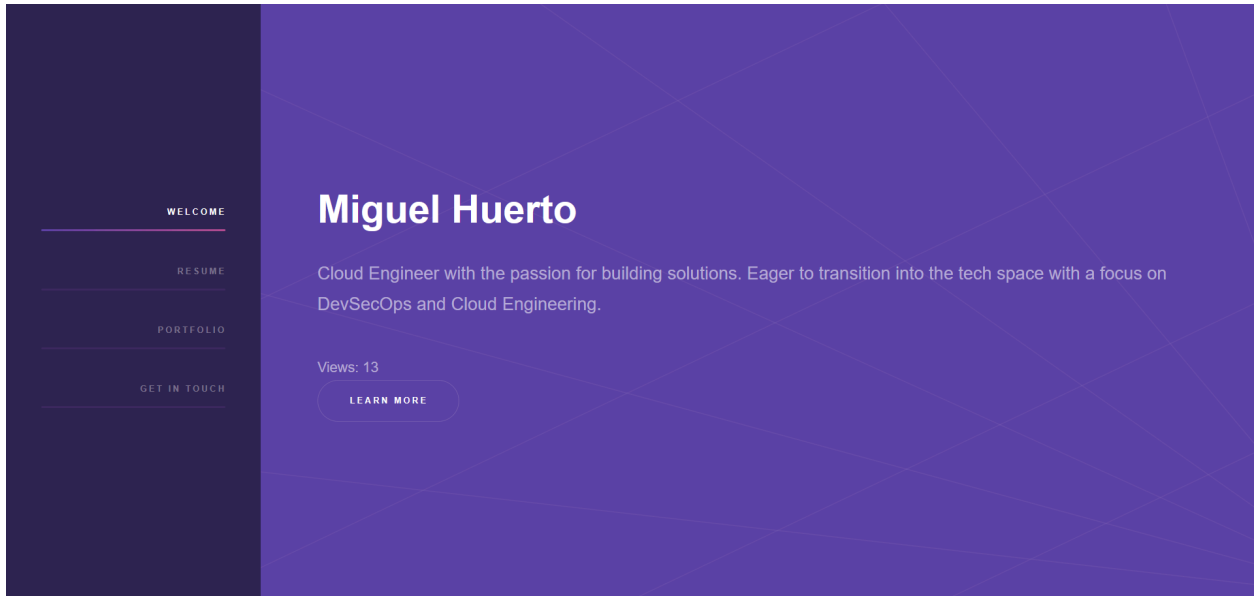
3. HTTPS



4. DNS

5. Javascript





6. Database

Tables (1) Info

Find tables

Any tag key

Any tag value

< 1 >

⚙️

☐	Name	Status	Partition key	Sort key	Indexes	Replication Regions	Deletion protection	Favorite	Read cap
☐	cloud-resume-test	Active	id (S)	-	0	0	Off	☆	On-dem

Tables (1)

Any tag key

Any tag value

Find tables

< 1 >

⚙️

cloud-resume-test

☆

cloud-resume-test

Autopreview

View table details

▶ Scan or query items

Expand to query or scan items.

✔ Completed. Read capacity units consumed: 2

×

Items returned (1)

⌂

Actions

Create item

< 1 >

⚙️

🔍

☐	id (String)	views
☐	1	1

7. API

Routes

Stage: - ▾

Deploy

Routes for cloud-challenge-resume

Create

Search

▼ /cloud-resume-test-api

ANY

Route details

ANY /cloud-resume-test-api (ID: h66zqbh)

Delete

Edit

ARN

arn:aws:apigateway:us-east-1::apis/kbat9glkqb/routes/h66zqbh

Authorization

Authorizers protect your API against unauthorized requests. Routes with no authorization attached are open.

No authorizer attached to this route.

Attach authorization

Integration

The integration is the backend resource that this route calls when it receives a request.

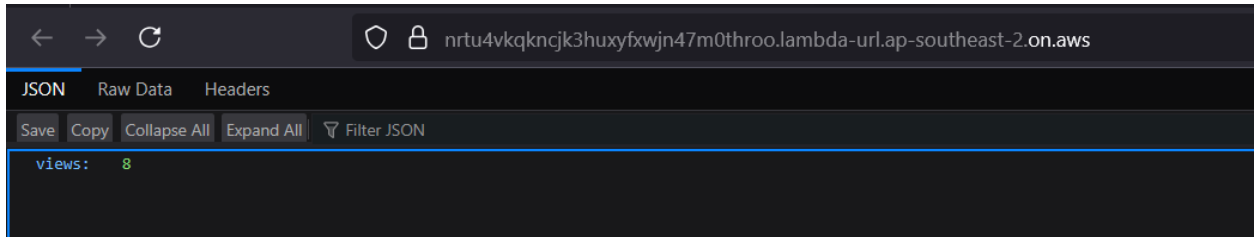
sjrosbf

Configure

8. Python

lambda_function.py

```
1 import json
2 import boto3
3
4 dynamodb = boto3.resource('dynamodb')
5 table = dynamodb.Table('cloud-resume-test')
6
7 def lambda_handler(event, context):
8     response = table.get_item(Key={'id': '0'})
9     views = response['Item']['views']
10    views += 1
11
12    table.update_item(
13        Key={'id': '0'},
14        UpdateExpression="SET #v = :val",
15        ExpressionAttributeNames={"#v": "views"},
16        ExpressionAttributeValues={":val": views},
17        ReturnValues="UPDATED_NEW"
18    )
19
20    return {"views": views}
21
```



9. Infrastructure as Code (TerraForm)

```
cloud-resume-challenge > infrastructure > provider.tf > ...
1  terraform {
2    required_providers {
3      aws = {
4        source  = "hashicorp/aws"
5        version = "~> 5.0"
6      }
7    }
8  }
9
10 provider "aws" {
11   region = "ap-southeast-2"
12   profile = "default"
13 }
14
```

```

cloud-resume-challenge > infrastructure > main.tf > resource "aws_iam_role" "iam_for_lambda"
1  resource "aws_s3_bucket" "resume-bucket" {
2      bucket = "cloud-resume-copy"
3  }
4
5  resource "aws_s3_bucket_ownership_controls" "bucket-ownership" {
6      bucket = aws_s3_bucket.resume-bucket.id
7
8      rule {
9          object_ownership = "ObjectWriter"
10     }
11 }
12
13 resource "aws_s3_bucket_acl" "resume-bucket-acl" {
14     bucket = aws_s3_bucket.resume-bucket.id
15     acl    = "public-read"
16     depends_on = [aws_s3_bucket_ownership_controls.bucket-ownership]
17 }
18
19 resource "aws_lambda_function" "myfunc" {
20     filename      = data.archive_file.zip_the_python_code.output_path
21     source_code_hash = data.archive_file.zip_the_python_code.output_base64sha256
22     function_name = "myfunc"
23     role          = aws_iam_role.iam_for_lambda.arn
24     handler       = "func.handler"
25     runtime       = "python3.8"
26 }
27
28 resource "aws_iam_role" "iam_for_lambda" {
29     name = "iam_for_lambda"
30
31     assume_role_policy = <<EOF
32 {
33     "Version": "2012-10-17"

```

10. Source Control

The screenshot shows the GitHub interface for a repository named 'cloud-resume-challenge'. At the top, the repository name is displayed with a 'Public' badge and buttons for 'Pin' and 'Unwatch'. Below this, the branch 'main' is selected, showing '1 Branch' and '0 Tags'. A search bar 'Go to file' and buttons 'Add file' and 'Code' are visible. The commit history shows an 'Initial Commit' by user 'miguelhhuerto' with commit hash '66eaf3d' from '4 minutes ago'. Below the commit, a file named 'cloud-resume-challenge' is listed as part of the 'Initial Commit' from '4 minutes ago'.

11. CI/CD (GitHub Actions)

← Upload website to S3

✓ Fix Front-end CICD #2 Re-run all jobs Latest #4 ...

Summary

Jobs

- ✓ deploy

Run details

Usage

Workflow file

Re-run triggered 3 minutes ago

Status: Success

Total duration: 24s

Artifacts: -

front-end-cicd.yaml

on: push

✓ deploy 20s

```
! front-end-cicd.yaml X Od00666 - Fix Front-end CICD (1 file)

.github > workflows > ! front-end-cicd.yaml
1  name: Upload website to S3
2
3  on:
4    push:
5      branches:
6        - main
7
8  jobs:
9    deploy:
10     runs-on: ubuntu-latest
11     steps:
12       - name: Checkout repository
13         uses: actions/checkout@v4
14
15       - name: Sync files to S3
16         uses: jakejarvis/s3-sync-action@master
17         with:
18           args: --acl public-read --follow-symlinks --delete
19         env:
20           AWS_S3_BUCKET: ${ secrets.AWS_S3_BUCKET }
21           AWS_ACCESS_KEY_ID: ${ secrets.AWS_ACCESS_KEY_ID }
22           AWS_SECRET_ACCESS_KEY: ${ secrets.AWS_SECRET_ACCESS_KEY }
23           AWS_REGION: 'ap-southeast-2'
24           SOURCE_DIR: 'cloud-resume-challenge'
25
```